```
% Analog signal to Digital signal
clear all
clc
Fs = 150;
T=1/Fs;
bit=2;
dt = 1/Fs;
StopTime = 0.025;
t=0:0.00001:StopTime;
n=0:1:(StopTime/dt);
Fc =100;
Ya =5*sin(2*pi*Fc*t)+3*cos(2*pi*6*Fc*t);
y_max=max(Ya);
y_min=min(Ya);
y =5*sin(2*pi*(Fc/Fs)*n)+3*cos(2*pi*6*(Fc/Fs)*n);
ns=length(y);

q_out=zeros(1,ns);
del=(y_max-y_min)/(2^bit);
low=y_min+(del/2);
high=y_max-(del/2);

for h=low:del:high
  for b=1:ns
    if(((h-del/2)<y(b)) && ((y(b)<=(h+del/2))))
      q_out(b)=h;
    end
  end
end

q_error=q_out-y;
```

```
r_memory=bit*ns;
fprintf('Required memory: %d bit',r_memory);


subplot(321)
plot(t,Ya,'r','Linewidth',1);
xlabel('Time');
ylabel('amplitude');
title('Analog Signal');


subplot(322)
stem(n,y,'c','Linewidth',1);
xlabel('sample number');
ylabel('amplitude');
title('Discrete time continuous valued Signal');


subplot(323)
stem(n,q_out,'m','Linewidth',1);
xlabel('Sample number');
ylabel('amplitude');
title('Digital (discrete time discrete valued) Signal');


subplot(324)
plot(n,q_error,'b','Linewidth',1);
xlabel('Sample Number');
ylabel('amplitude');
title('Quantization Noise');


subplot(325)
plot(n.*T,q_out,'g','Linewidth',1);
xlabel('Time');
ylabel('amplitude');
```

```
title('Reconstructed Analog Signal');


subplot(326)

plot(n.*T,q_out,'g','Linewidth',1);

hold on

plot(t,Ya,'r','Linewidth',1);

title('Discrete time continuous valued Signal');

xlabel('Time');

ylabel('amplitude');

title('Reconstructed Analog Signal Vs Analog Signal');
```

```
%Shifting

clc;

clear;

close all;

%Original signal

x=[1 3 5 7 9];

n=-2:2;

%Shifting amount

k=2;

%Left shift: y[n]=x[n+k]

n_left=n-k;

%Right shift: y[n]=x[n-k]

n_right=n+k;

%Plot signal

figure;

%Original signal

subplot(3,1,1);

stem(n,x,'filled','LineWidth',1.5);

grid on;

title('Original Signal x[n]');
```

```
xlabel('n');
ylabel('Amplitude');
%Left shift
subplot(3,1,2);
stem(n_left,x,'filled','LineWidth',1.5);
grid on;
title('Left Shift by 2:x=[n+2]');
xlabel('n');
ylabel('Amplitude');
%Right Shift
subplot(3,1,3);
stem(n_right,x,"filled",'LineWidth',1.5);
grid on;
xlabel('n');
ylabel('Amplitude');
```

```
%Folding
clc;
clear;
close all;
%Original Signal
x=[1 3 5 7 9];
n=-2:2;
%Folding(Time Reversal)
y=fliplr(x);
n_fold=-fliplr(n);
%Plot Original Signal
subplot(2,1,1);
stem(n,x,'filled','LineWidth',1.5);
grid on;
title('Original Signal x[n]');
```

```
xlabel('n');
ylabel('Amplitude');
%Plot Folded Signal
subplot(2,1,2);
stem(n_fold,y,'filled','LineWidth',1.5);
grid on;
title('Folded Signal y[n]=x[-n]');
xlabel('n');
ylabel('Amplitude');
```

```
%Even and Odd Part
clc;
clear;
close all;
%Original Signal
x=[2 4 6 8 10];
n = -2:2;
%Folded signal x[-n]
x_fold = fliplr(x);
%Even and Odd part
x_even = (x+x_fold)/2;
x_odd = (x-x_fold)/2;
%Display values
disp('original Signal x[n]=');
disp(x);
disp('Folded Signal x[-n]=');
disp(x_fold);
disp('Even part x_e[n]=');
disp(x_even);
disp('Odd part x_o[n]=');
disp(x_odd);
```

```
%Plotting
figure;
%Original Signal
subplot(2,2,1);
stem(n,x,'filled','LineWidth',1.5);
grid on;
title('Oringinal Signal x[n]');
xlabel('n');
ylabel('Amplitude');
%Folded Signal
subplot(2,2,2);
stem(n,x_fold,'filled','LineWidth',1.5);
grid on;
title('Folded Signal x[-n]');

xlabel('n');
ylabel('Amplitude');
%Even part
subplot(2,2,3);
stem(n,x_even,'filled','LineWidth',1.5);
grid on;
title('Even Part x_e[n]');
xlabel('n');
ylabel('Amplitude');

%Odd Part
subplot(2,2,4);
stem(n,x_odd,'filled','LineWidth',1.5);
grid on;
title('Odd Part x_o[n]');
xlabel('n');
```

```
ylabel('Amplitude');
```

....

```
%Convolution
clc;
clear;
close all;
% Input signals
x = [1 2 3];
h = [1 2 1];
% Perform convolution
y = conv(x, h);

% Display results
disp('Signal x[n] =');
disp(x);
disp('Signal h[n] =');
disp(h);
disp('Convolution y[n] = x[n] * h[n]');
disp(y);
% Define indices
nx = 0:length(x)-1;
nh = 0:length(h)-1;
ny = 0:length(y)-1;
% Plot signals
figure;
% Input signal x[n]
subplot(3,1,1);
stem(nx, x, 'filled', 'LineWidth', 1.5);
grid on;
title('Input Signal x[n]');
xlabel('n');
```

```
ylabel('Amplitude');
% Input signal h[n]
subplot(3,1,2);
stem(nh, h, 'filled', 'LineWidth', 1.5);
grid on;
title('Input Signal h[n]');
xlabel('n');
ylabel('Amplitude');
% Convolution result
subplot(3,1,3);
stem(ny, y, 'filled', 'LineWidth', 1.5);
grid on;
title('Convolution Result y[n] = x[n] * h[n]');
xlabel('n');
ylabel('Amplitude');
```

```
%Correlation
clc;
clear;
close all;
% Input signals
x = [1 2 3];
y = [1 2 1];
% Cross-correlation
r = xcorr(x, y);
% Display results
disp('Signal x[n] =');
disp(x);
disp('Signal y[n] =');
disp(y);
disp('Cross-Correlation r_xy[l] =');
```

```
disp(r);

% Define indices

nx = 0:length(x)-1;

ny = 0:length(y)-1;

l = -(length(y)-1):(length(x)-1);

% Plot signals

figure;

% Signal x[n]

subplot(3,1,1);

stem(nx, x, 'filled', 'LineWidth', 1.5);

grid on;

title('Signal x[n]');

xlabel('n');

ylabel('Amplitude');

% Signal y[n]

subplot(3,1,2);

stem(ny, y, 'filled', 'LineWidth', 1.5);

grid on;

title('Signal y[n]');

xlabel('n');

ylabel('Amplitude');

% Cross-Correlation

subplot(3,1,3);

stem(l, r, 'filled', 'LineWidth', 1.5);

grid on;

title('Cross-Correlation r_{xy}[l]');

xlabel('Lag (l)');

ylabel('Amplitude');
```